MODULE 1 PROMPT ENGG. & API INT.

SESSION 6: Advanced Handling of LLM Responses

Lang Chain

1. Chain multiple LLM steps.

1000s Extract text ⟶ Summarize. ⟶ Match Jobs.

2. Maintain Memory / context

3. Using tools with LLMs

4. Q&A From Documents

5. Scaling: Batch / Async / streaming.

Capabilities of LangChain

① LLM Integration - Need to support most if not all LLMs that are Frequently used

② Prompt Management - Prompt Template to provide Reusable and Dynamic Instructions.

③ Chains multiple steps

④ Memory / caching.

⑤ Tools to add capabilities

⑥ Agents - Decide what to do?

⑦ Retrieve Data

⑧ Streaming outputs. / Batch processing.

⑨ Deploy Apps. - LangServe

Handling failures

1. Try Except
2. Logging errors
3. Retry.
4. Fallback.
5. Post processing.

Guardrails

Needed because

1. LLMs are probabilistic with no guarantee of correctness
2. Context sensitive.
3. Prompt Injection
4. Hallucinates
5. Bias / Unsafe content can be exposed.
6. Decision Making is done by bots.

How do Guardrails work in LLM Applications?

Checker Corrector. Rail Guard.

The watchdog. Fixer Conductor Orchestrator.